

MATHEMATICAL GAMES

Magic stars, graphs and polyhedrons, and answers to last month's problems

by Martin Gardner

"I've always been poor at geometry,"
he began...

"You're telling me," said the demon
gleefully.

Smiling flames, it came for him across
the chalk lines of the useless hexagram
Henry had drawn by mistake instead of
the protecting pentagram.

—FREDRIC BROWN,
from "Naturally," in *Honeymoon in Hell*

The field of combinatorial arithmetic has been getting increasing attention from mathematicians during the past decade, and along with this revival has come a new interest in combinatorial problems that were once considered mere puzzles. Herbert J. Ryser begins his excellent little book *Combinatorial Mathematics* (published in 1963 by the Mathematical Association of America) by displaying the three-by-three magic square, which was known in China centuries before the Christian era. "Many of the problems studied in the past for their amusement or aesthetic appeal are of great value today in pure and applied science," he writes. "Not long ago finite projective planes were regarded as a combinatorial curiosity. Today they are basic in the

foundations of geometry and in the analysis and design of experiments. Our new technology with its vital concern with the discrete has given the recreational mathematics of the past a new seriousness of purpose."

Magic squares were briefly introduced in this department in March, 1959. This month I turn to the less widely known but closely related topic of magic stars. It is a branch of recreational combinatorics that has a fascinating overlap with graph theory and the skeletal structure of polyhedrons. It is also rich in problems.

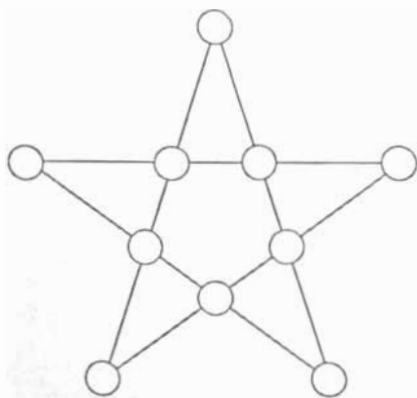
The simplest star polygon is the familiar five-pointed Christmas star, which as children we learned to draw in one continuous path of five straight lines. It served as a recognition symbol for the ancient Greek Pythagoreans, and it was also their symbol of health. In medieval and Renaissance witchcraft it was the mystic "pentagram" or "pentalfa." (The second name derives from the fact that it can be formed by superposing five capital A's.) The three large isosceles triangles that can also be superposed to make the star were taken as symbols of the Trinity and the star's points were often labeled J-E-S-U-S. When Goethe's Faust draws a pentagram on the threshold of his study, he fails to close the path. This slight break at one of the outer points allows Mephistopheles to enter the room, only to find himself trapped by the closed curve of the star's inner pentagon. Later, while Faust sleeps, the demon escapes by ordering a rat to nibble an opening in this pentagon.

Make a circle at each vertex of a pentagram [see illustration at left]. Is it possible to put the integers 1 through 10 in these 10 circles so that each line of four numbers will have the same sum? It is easy to determine what this sum, the "magic constant," must be. The numbers 1 through 10 sum to 55. Each number appears in *two* lines, therefore the sum of all five line sums must be twice 55, or 110. Since the five line sums are equal, each line must have a sum of $110/5$, or 22. If a magic penta-

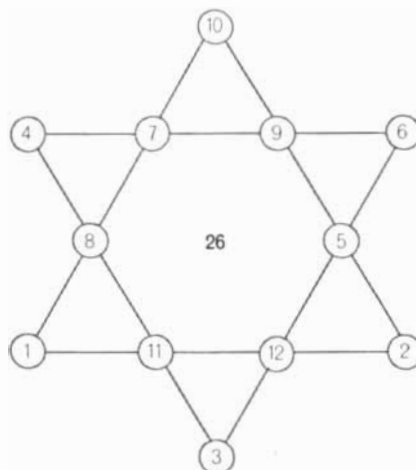
gram exists, therefore, its magic constant must be 22.

The fact that no such magic pentagram appears in the literature of witchcraft is strong evidence for its impossibility, and with a little ingenuity one can indeed prove that it cannot be accomplished. (Known proofs are lengthy, and so I must refer the interested reader to Harry Langman's *Play Mathematics*, published in 1962, in which a proof is given.) The best we can do—without duplicating a number or using zero or negative numbers—is to label the vertices with 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, as shown at the left in the upper illustration on the opposite page. This makes a defective magic pentagram with the lowest possible constant, 24, and the lowest high number, 12.

Consider now the following question, seemingly unrelated to the pentagram. Is it possible to label the 10 edges of the pentatope, or four-dimensional tetrahedron, so that at each corner the sum of the edges meeting at that corner is the same? Surprisingly, the question has just been answered; in its combinatorial aspect it is identical with the question about the pentagram! First draw the graph shown in the middle of the upper illustration on the opposite page. This is called the "complete graph" for five points, because it joins each point to all the others. If you compare the numbers at the vertices of the pentagram with those on the edges of the graph, you will see that there is an identity of combinatorial structure. Every line of four numbers in the star matches a cluster of four numbers on edges meeting at a common point. Because the magic star is impossible, it is impossible to make the complete graph



The pentagram



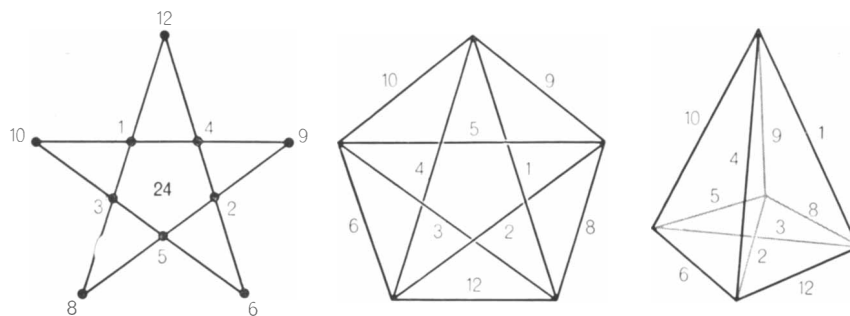
Magic hexagram

for five points “magic” at its vertices. Now, the complete graph for five points is topologically the same as the skeleton of the four-dimensional tetrahedron, as you can verify by comparing the numbers on the graph with those on a projection in three-space of the pentatope’s skeleton [at right in illustration]. A pentatope that is magic at the vertices is therefore not possible. Since the numbers shown on the pentatope’s skeleton map back to those on the pentagram, we know that we have provided the pentatope with a nonconsecutive solution that has the lowest constant and lowest high number.

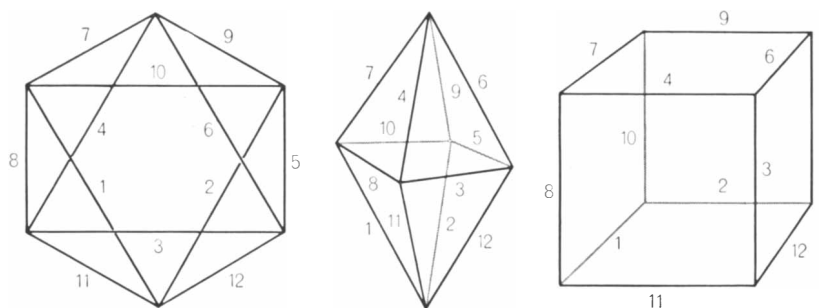
The situation grows in interest when we turn to the hexagram—also known as the hexalpha, Solomon’s seal and the Star of David [see illustration at bottom right on opposite page], a figure almost as prominent in the history of occultism and superstition as the pentagram. Because there are six lines, with each vertex common to two lines, and because the numbers 1 through 12 sum to 78, we obtain the magic constant $(2 \times 78)/6$, or 26. As the illustration shows, a magic hexagram is possible.

The problem of cataloguing all the different hexagram solutions, not counting rotations and reflections as being different, is not easy. One way to obtain new patterns is to transform the hexagram to its dual graph [at left in lower illustration on this page], on which the same numbers mark lines that are magic at the vertices. It is easy to see that this graph is topologically the same as the skeleton of the octahedron [middle], one of the five Platonic solids. We can now rotate the octahedron and mirror-reflect it in any way we please, then map the numbers back to the hexagram (mapping edges to vertices according to the original numbering) and obtain new patterns for the hexagram. Other transformations of the hexagram can be made, unrelated to rotations and reflections of the octahedron, that give still more solutions. Moreover, every magic star has what is called its “complement,” obtained by replacing each number with the difference between that number and $n + 1$, when n is the highest of the star’s consecutive integers. There are 74 different solutions, 12 with outer star points that also sum, as in the solution shown, to the constant.

There is still more to be said: the octahedron has what is called its “polyhedral dual,” in which every face is replaced by a vertex and every vertex by a face, the edges remaining the same. The octahedron’s dual is the cube. This allows us to label the 12



Pentagram (left) and equivalent graph (middle) and pentatope (right)



Graph of hexagram (left) and equivalent octahedron (middle) and cube (right)

edges of a cube [at right in illustration] with numbers 1 through 12 so that the cube is magic at its faces; that is, the sum of the four edges bounding every face is 26.

Can the septagram, or seven-pointed star [top left on next page], be made magic by labeling its vertices with numbers 1 through 14? Yes, and I leave it to the reader to see how quickly he can find one of its 56 different solutions. The constant is $(2 \times 105)/7$, or 30. The best way to work on it is to draw a large figure, then put the numbers on small counters that can be slid over the paper. Warning: Once you start, you will find it hard to stop until you get a solution. An answer will be given next month.

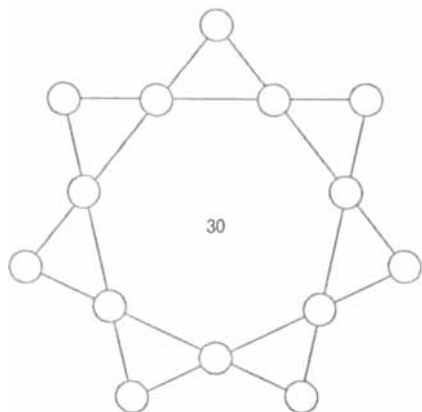
One solution for the octagram, or eight-pointed star, is shown at the top of the bottom illustration on the next page. Note that the magic constant, 34, is also the sum of the four corners of each of the two large squares. The middle drawing shows a corresponding graph that is magic at its vertices, and the drawing at the bottom shows a solid with an equivalent skeleton. It has not yet been determined, as far as I know, how many solutions there are for the octagram.

Clearly there is no end to combinatorial problems that have to do with the labeling of edges, vertices or faces of various polyhedrons so that magic constants are obtained in various ways. Many of these problems translate to

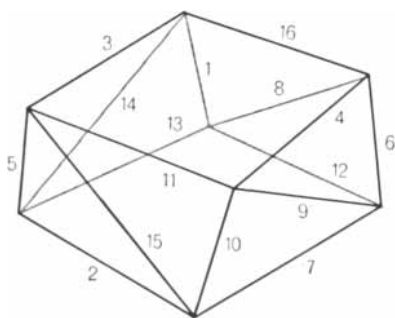
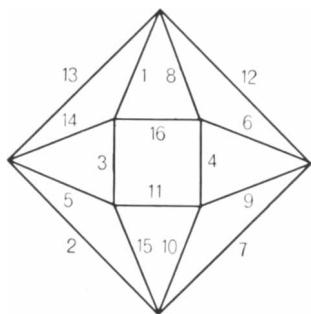
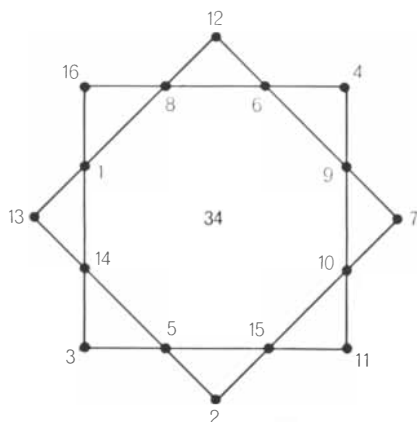
equivalent magic star problems. For example, which of the five regular solids can be made magic at their corners by being labeled along their edges with consecutive integers? It is easy to show that this is not possible on the tetrahedron. (See this department for last March.) Is it possible on the cube? The cube’s 12 edges [top of illustration at top right on next page], map to the 12 black spots on the vertices of the octagram [bottom]. Since each spot is in two lines, the constant must be $(2 \times 78)/8$, or $19\frac{1}{2}$. This is not an integer, so that we know at once the problem has no solution. The best we can do is to label the spots (or the cube’s edges) as shown, to get a defective solution with the lowest constant, 20, and the lowest high number. Since the octahedron is the polyhedral dual of the cube, this automatically solves the problem of labeling the octahedron’s edges with different nonconsecutive, nonzero, positive integers to obtain the lowest constant at the faces.

We have seen that the octahedron’s edges can be labeled with consecutive integers to make it magic at its corners. On the icosahedron and dodecahedron the constant is not integral, so that they have no solutions. Since each is the other’s polyhedral dual, there are no solutions for the corresponding problems of making them magic at their faces.

If we mark only nine of the hexagram’s vertices as shown by the spots at



The septagram



Octagram and two equivalents

the left in the top illustration on the opposite page, we obtain a magic star problem equivalent to that of labeling the nine edges of a triangular prism [at right] with numbers 1 through 9 to make it magic at its six corners. The constant must be $(2 \times 45)/6$, or 15. Can it be done? It is not a difficult question, and next month I shall give either a proof of impossibility or a solution.

The answers to the collection of short word and number problems presented last month follow.

1.

Dmitri Borgmann, author of *Language on Vacation*, is my authority for the following answers to the questions about the anagram dictionary.

1. Among common noncolloquial words, *sty*, *sty*, is probably the last entry. But this is followed by *ttu*, *tut*, *ttuv*, *tutu* (a short ballet skirt), *tux*, *tux* (short for tuxedo) and *zzzz*, *zzzz* (to snore), which Borgmann says is in the second edition of *The American Thesaurus of Slang*, by Lester V. Berrey and Melvin Van Den Bark.

2. The first and second entries are *A*, *a*, and *AA*, *aa* (a kind of lava).

3. The last entry is *ay*, unless we accept *ayy*, *yay* (an obsolete variant of "they").

4. Among common words, the first entry beginning with *B* is *BBBDEI*, *bibbed*. It is preceded by the less common *BBBBBEEHLLUV*, hubble-bubble (a bubbling sound, also a hookah).

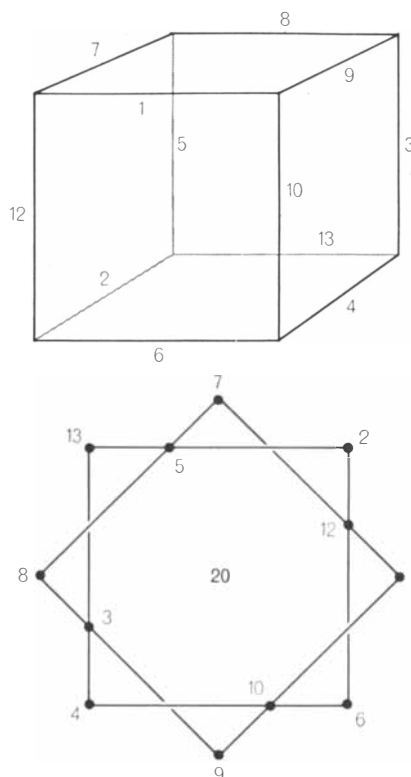
5. *ABCDEFOL*, *boldface*.

6. The longest entry that is itself a common word (that is, a word with letters in alphabetical order) is *billowy*. In *Language on Vacation* Borgmann supplies a longer word, *aegilops* (a genus of grasses).

7. The longest common English word that does not repeat a letter is *uncopyrightables*. But Borgmann supplies some longer coined words, such as *vodka-thumbscrewingly*, with 20 letters, which means to apply thumbscrews while under the influence of vodka. The longest such word ever created, he says, is the 23-letter monster *pubvexingfjordschmaltzy*, which means "as if in the manner of the extreme sentimentalism generated in some individuals by the sight of a majestic fjord, which sentimentalism is annoying to the clientele of an English inn."

2.

The girl's escape strategy is first to



Magic cube skeleton (top) and octagram

row so that the lake's center, marked by the raft, is always between her and the man on shore, the three points maintaining a straight line. The man runs four times faster than she can row, so that she can do this, and at the same time move shoreward, until she reaches a distance from the center that is a quarter of the radius of the lake. That is the point at which the angular velocity she must maintain to keep him opposite her just equals his, leaving her no reserve energy to use in moving outward. As soon as she reaches this point she heads straight for the nearest spot on shore. She has a distance of $3r/4$ to go (r is the lake's radius), whereas he has to travel a distance of πr to catch her when she lands. Because he travels four times faster than she rows, by the time she has reached the shore he has gone a distance of only $3r$. This is less than πr , and the result is that he misses her.

3.

To transfer the key from one side of the door to the other, first pass the key through the loop so that it hangs as shown at the left in the bottom illustration on the opposite page. Seize the double cord at points *A* and *B* and pull the loop back through and out of the keyhole. This will pull two new loops

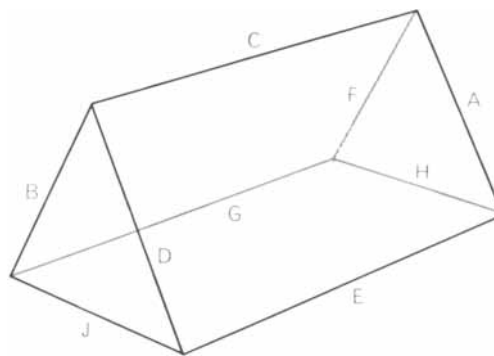
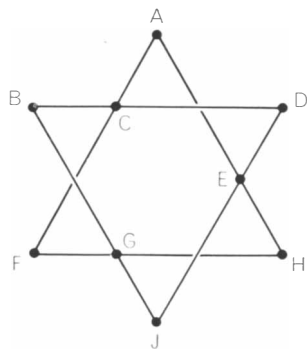
out of the hole, as shown in the middle. Move the key up along the cord, through both projecting loops. Grasp the two cords on the opposite side of the door and pull the two loops back through the hole, restoring the cord to its original state [right]. Slide the key to the right, through the loop, and the job is done.

4.

The smallest number of unit line segments that can be removed from a four-by-four checkerboard to render it "square-free" is nine. One way to do this is shown on the interior four-by-four square at the top of the top illustration on the next page.

To prove this minimum, note that the eight shaded cells have no side in common; to break the perimeters of all eight at least eight unit lines must be removed. The same argument applies to the eight white cells. We can, however, "kill" all 16 cells with the same eight lines if we pick lines shared by adjacent cells so that each erased line kills a white and a shaded cell simultaneously. But if we do this, not one of the removed lines will be on the board's outside border, which forms the largest square. Therefore at least nine cells must be removed to kill the 16 small cells plus the outer border. As the solution shows, the same nine will eliminate all 30 squares on the board.

The same argument proves that every even-order square must have a solution at least equal to $\frac{1}{2}n^2 + 1$, n being the square's order. Can this be achieved on all even-order squares? A



Is a magic triangular prism (right) possible? Equivalent star form is at left

proof by induction is implicit in the procedure shown in the illustration. We merely plug a domino in the open cell on the border of the four-by-four, then run a chain of dominoes around the border as shown. This provides a minimum solution of 19 for the order-six board. The same procedure is applied once more to give the minimum solution of 33 for the eight-by-eight board. It is obvious that this procedure can be repeated endlessly, with each new border of dominoes raising the open cell one step, as shown by the arrow.

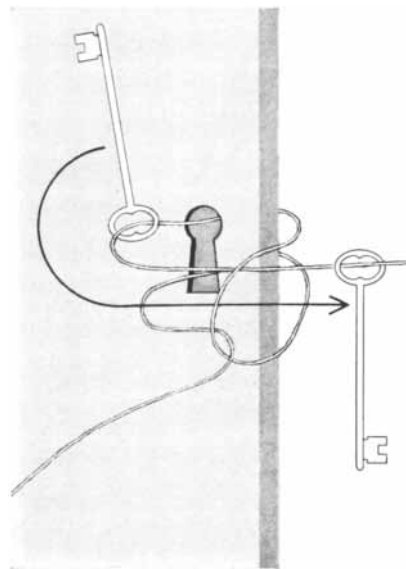
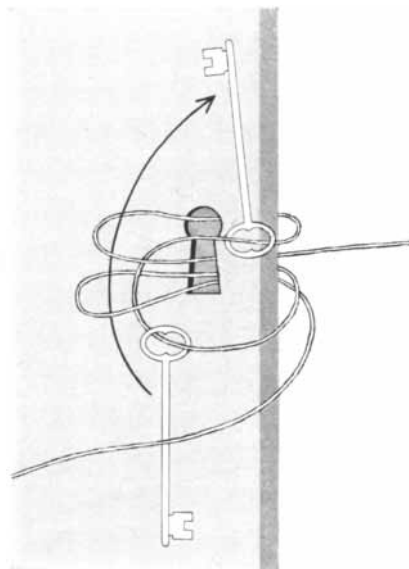
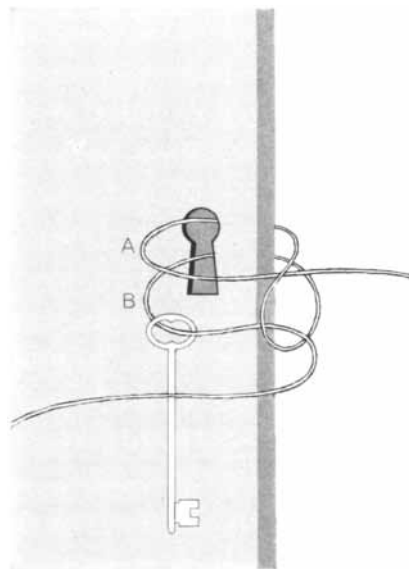
On the order-five board the situation is complicated by the fact that there is one more shaded cell than there are white cells. At least 12 lines would have to be removed to kill 12 shaded and 12 white cells simultaneously. This would, of course, form 12 dominoes. If the remaining shaded cell were on the outside border, both this cell and the border could be killed by taking one more line, which suggests that odd-order squares might have a minimum so-

lution of $\frac{1}{2}(n^2 + 1)$. To achieve this, however, the dominoes would have to be arranged so as not to form an unbroken square higher than order-one. It can be shown that this is never possible, so that the minimum is raised to $\frac{1}{2}(n^2 + 1) + 1$. The lower drawing in the top illustration on the next page shows a procedure that achieves this minimum for all odd-order squares.

5.

It is easy to show that for any finite set of points on the plane there must be an infinity of straight lines that divide the set exactly in half. The following proof for the six points in the bottom illustration on the next page applies to any finite number of points.

Consider every line determined by every pair of points. Pick a new point, A, that lies outside a closed curve surrounding all the other points and that does not lie on any of the lines. Draw a line through point A. As this line is



Solution to key-and-cord puzzle

The Biology of Viruses

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Spore Liberation

By C. T. INGOLD, *University of London*. Although this book deals with problems of spore liberation, particularly in fungi, it is not a new edition of the author's *Dispersal In Fungi*. The form and function of spores is described, with a full account of spore liberation in *Mucorales*, toadstools, *Gasteromycetes*, and *sordaria*. Rhythms of spore release are analyzed. A final chapter concerns bryophytes. The book's theme is that an understanding of the form of fungal fruit bodies and bryophytesporangia involves a consideration of spore liberation. 121 figures. \$5.60

Nitrogen Fixation in Plants

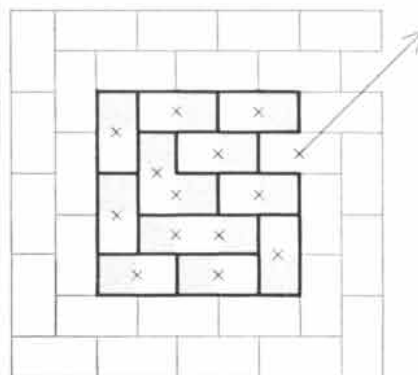
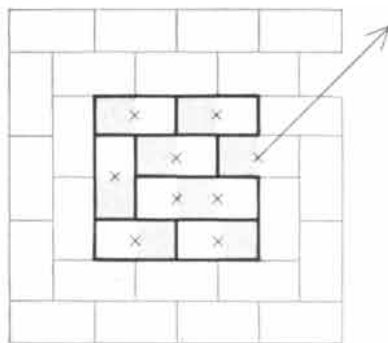
By W. D. P. STEWART. This work describes the physiology of nitrogen-fixing organisms and the biochemistry of the process; there has previously been no comprehensive up-to-date account of the subject. A unique feature of the book is an analysis of the ecology of nitrogen fixation, a field in which the author has carried out pioneer studies. 8 plates. \$6.40

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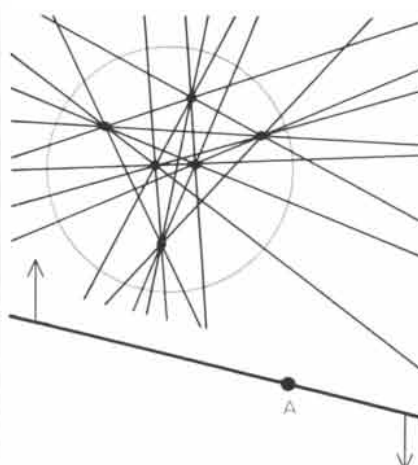


Solutions for toothpick problems

rotated about point A, in the direction shown, it must pass over one point at a time. (It cannot pass two points simultaneously; this would mean that point A lay on a line determined by those two points.) After it has passed half of the points inside the curve it will divide the set of points in half. Since A can be given an infinity of positions, there is an infinity of such lines.

6.

The first two statements can be satisfied only by two arrangements of Kings and Queens: KQQ and QKQ. The last



Proof for million-point problem

two statements are met by only two arrangements of Hearts and Spades: SSH and SHS. The two sets combine in four possible ways:

KS, QS, QH
KS, QH, QS
QS, KS, QH
QS, KH, QS

The last set is ruled out because it contains two Queens of Spades. Since each of the other three sets consists of the King of Spades, Queen of Spades and Queen of Hearts, we can be sure that those are the three cards on the table. We cannot know the position of any one card, but we can say that the first must be a Spade and the third a Queen.

7.

The three sets of four cocircular points on the picture of randomly dropped rectangles and disks are shown as black spots in the top illustration on the opposite page. The four corners of the rectangle were mentioned last month. The four points on the small circle are obviously cocircular. The third set consists of points A, B, C, D. To see this, draw dotted line BD and think of it as the diameter of a circle. Since the angles at A and C are right angles, we know (from a familiar theorem of plane geometry) that A and C must lie on the circle of which BD is the diameter.

8.

The most efficient procedure for testing any number of glasses of liquid in order to identify a single glass containing poison is a binary procedure. The glasses are divided as nearly in half as possible. One set is tested (by mixing samples from all the glasses and testing the mixture). The set known to include the poison glass is then again divided as nearly in half as possible, and the procedure is repeated until the poison glass is identified. If the number of glasses is from 100 to 128 inclusive, as many as seven tests might be required. From 129 to 200 glasses might take eight tests. The number 128 is the turning point, because it is the only number between 100 and 200 that is in the doubling series: 1, 2, 4, 8, 16, 32, 64, 128, 256. . . . There must have been 129 glasses in the hotel kitchen, because only in that case (we were told that the number was between 100 and 200) will the initial testing of one glass make no difference in applying the most ef-



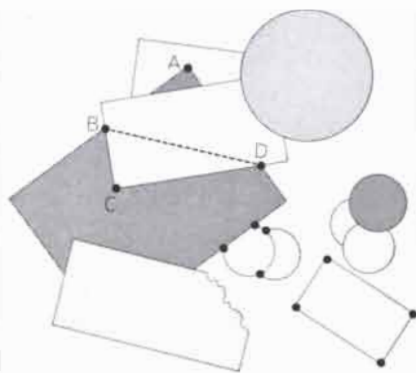
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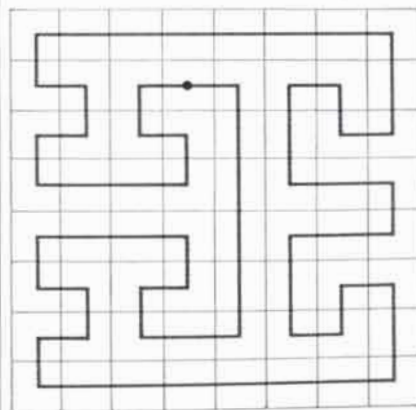
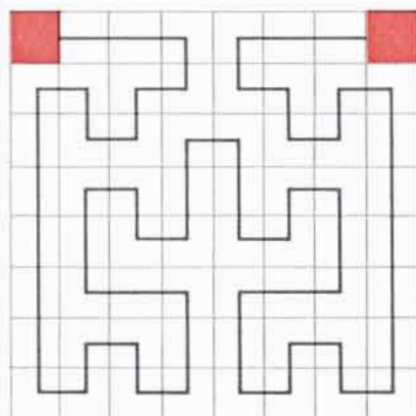


The three sets of cocircular points

ficient testing procedure. Testing 129 glasses, by halving, could demand eight tests. But if a single glass is tested first, the remaining 128 glasses require no more than seven tests, so that the total number of tests remains the same.

9.

Solutions to the cube-rolling tour problems are shown below. In the first solution the red side of the cube is up only on the top corner squares. In the second, the dot marks the start of the tour, with the red side down.



Cube-rolling solutions

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